Staining techniques & gram stain

Microbiology Lab

1rst year

College of Medicine

Lab. No. 4

Objectives:

1- Principle of staining.

2- Smear preparation

3- Types of staining techniques.

- Simple staining
- Differential staining (Gram staining)
- Special staining

4- Gram staining procedure

5- Acid fast staining prodedure

Principle of staining

- Bacterial cell are almost colorless and transparent .
- Stains combine chemically with the bacterial protoplasm. Their shape and size can be easily determined under the microscope.
- staining is almost always applied to color certain features of a specimen before examining it under a light microscope

- The commonly used stains are salts.

- Bacterial cells are rich in nucleic acid, bearing negative charges as phosphate groups. These combine with the positively charged basic dyes.

- Acidic dyes do not stain bacterial cells and hence can be used to stain background material a contrasting color.

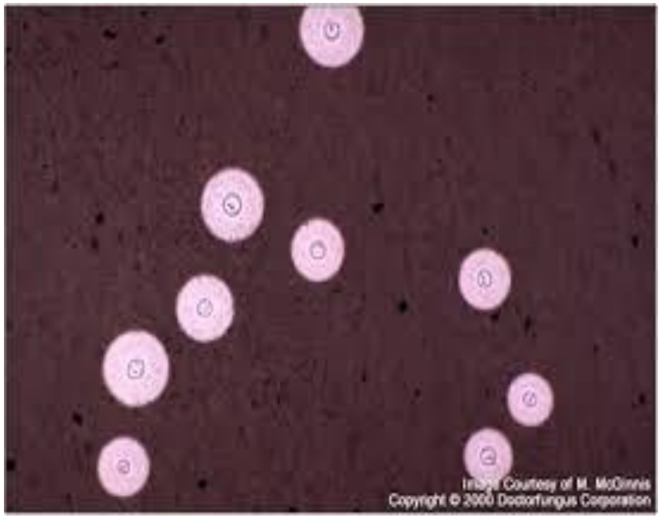
- Special staining techniques can be used to differentiate flagella, capsules, cell walls, cell membranes, granules, nucleoids, and spores.

NEGATIVE STAINING:

- simpler than simple stains because you do not have to make a smear.
- This procedure involves staining the background with an acidic dye, leaving the cells contrastingly colorless.

(a) Indian ink

(b) Nigrosin .



POSITIVE STAINING: Unlike negative staining, positive staining uses basic dyes to colour the specimen against a bright background.

(a) Simple staining: A stain which provides color contrast but gives same color to all bacteria and cells. Ex: Methylene blue, Diluted carbol fuchsin.

(b) Differential Staining: A stain which imparts different colors to different bacteria is called differential stain(which contains more than one stain).

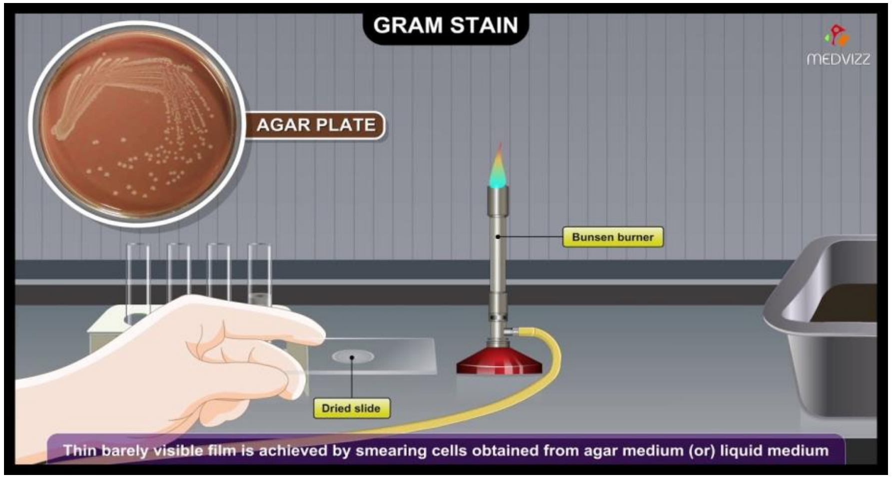
Ex: Gram's stain , Acid fast staining.

Preparation of bacterial smear

A Bacterial smear is a thin layer of bacteria placed on a slide for staining.

Purpose:

- kill the microorganism & fix them to the slide
- prevent them from being washed out during the process of staining.



TYPES OF STAINING TECHNIQUES

Simple staining

(use of a single stain)

For visualization of morphological shape & arrangement.

Differential staining

(use of two contrasting stains separated by a decolorizing agent)

Identification

- Gram stain
- Acid fast stain

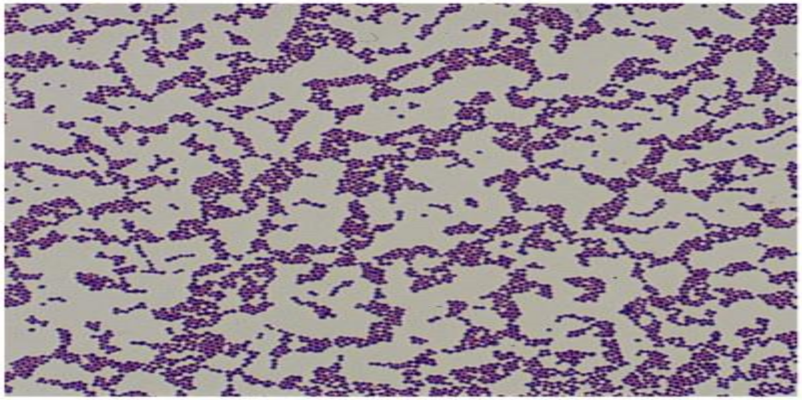
Simple staining

- It is the use of *single basic dye* to color the bacterial organism.

e.g. Methylene blue, crystal violet, safranin.

- Purpose:-

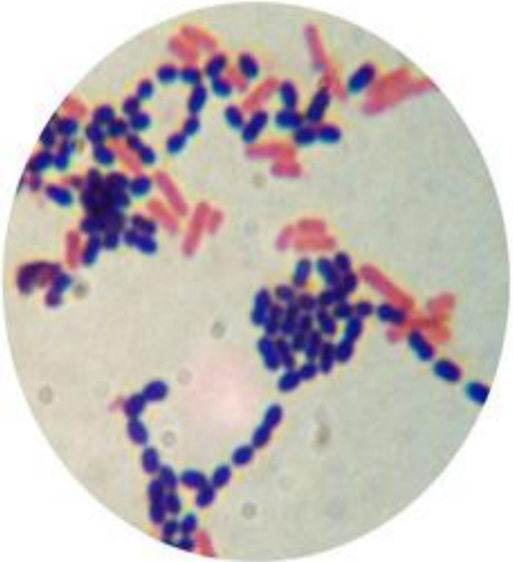
To show the morphological shapes and arrangement of bacterial cells.



(a) 24-hour culture

Differential stain

1- Gram Stain:



- It is the most important differential stain used in bacteriology because it classified bacteria into two major groups:

a)Gram positive:

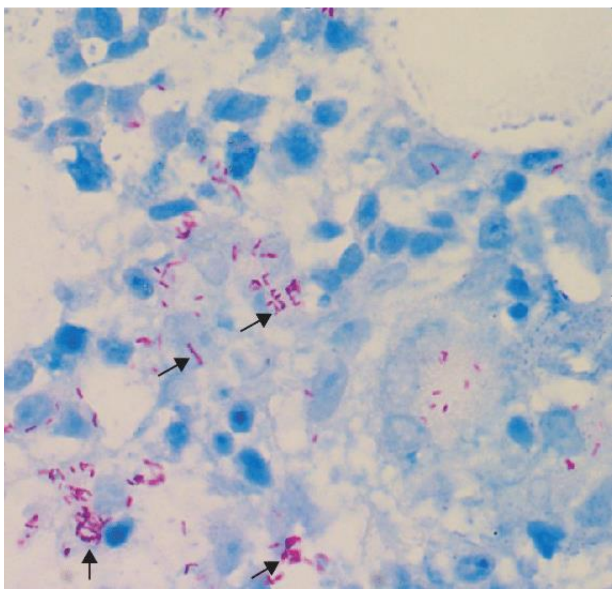
Appears **violet** after Gram's stain

b) Gram negative:

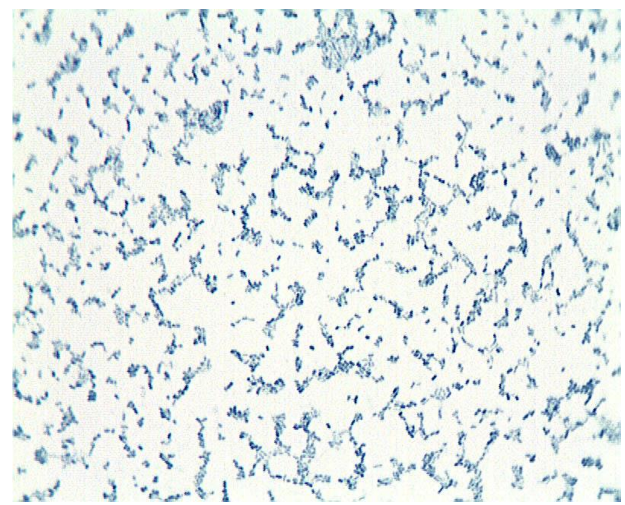
Appears **red** after Gram's stain

Differential stain
2- Acid fast Stain:

Bacteria that are still stained with red color after the decolorization step during the acid fast staining procedure are known as acid fast bacteria.



Acid Fast Bacteria



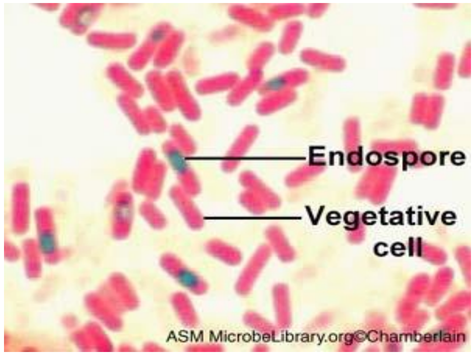
Non Acid Fast Bacteria

Special stain

- Used to stain special structures of bcteria :

Endospore stain

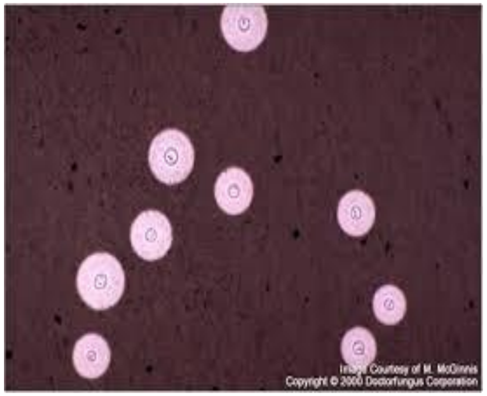
The presence of bacterial culture can be detected by staining with malachite green. Because the endospore coat is so tough, steam is used to **enable** dye penetration



Malachite green

Capsule stain

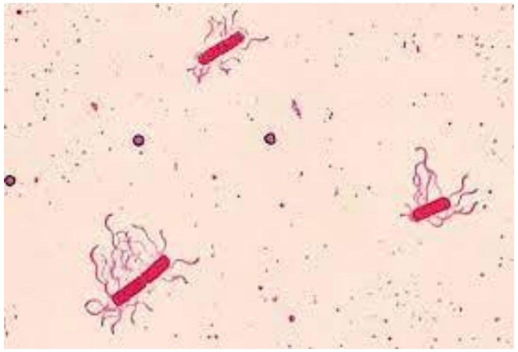
India ink capsule stain is used to demonstrate cell capsules through microscopic examination



India ink stain

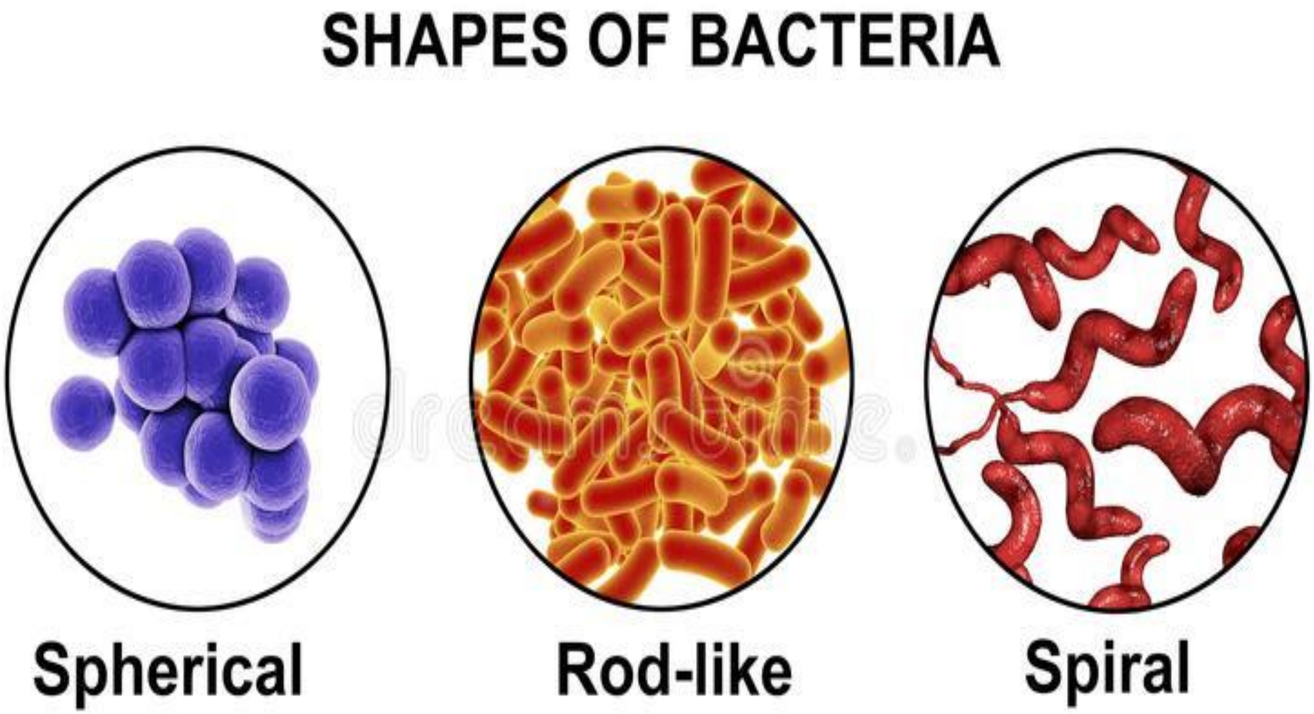
Flagella stain

The flagella stain allows observation of bacterial flagella under the light microscope.



Leifson flagella stain

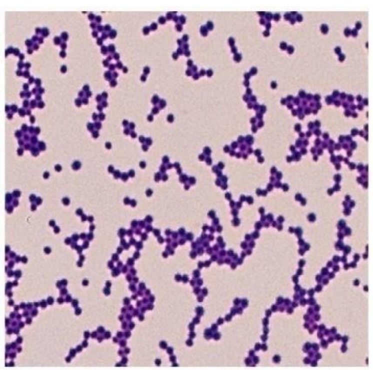
BASIC SHAPES OF BACTERIA



Arrangements

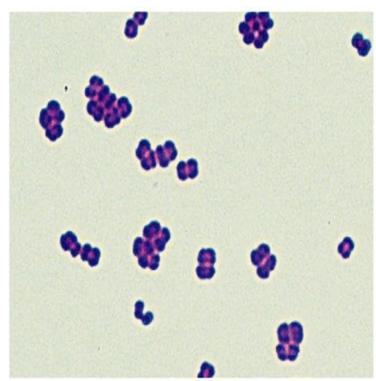
Cocci

- Irregular Clusters
- Tetrads
- Chains or Pairs



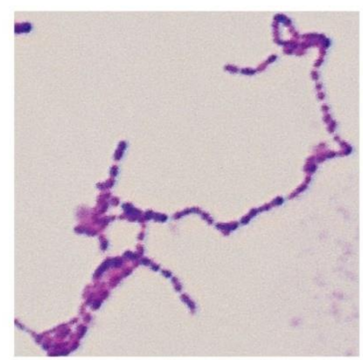
(e)

Staphylococci



(c)

Micrococci



(b)

Streptococci

Gram staining Procedure

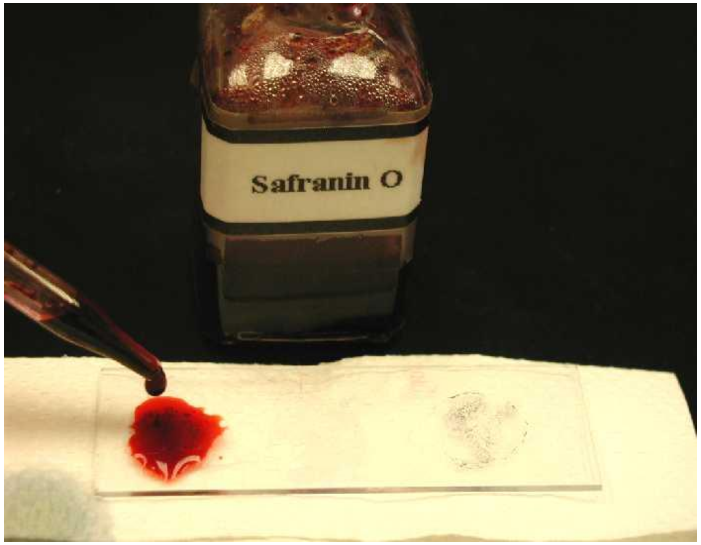
1. CRYSTAL VIOLET
   - Primary stain
   - Violet colored, stains all micro-organism

2. GRAM IODINE
   - Mordant
   - Forms Crystal violet iodine complexes

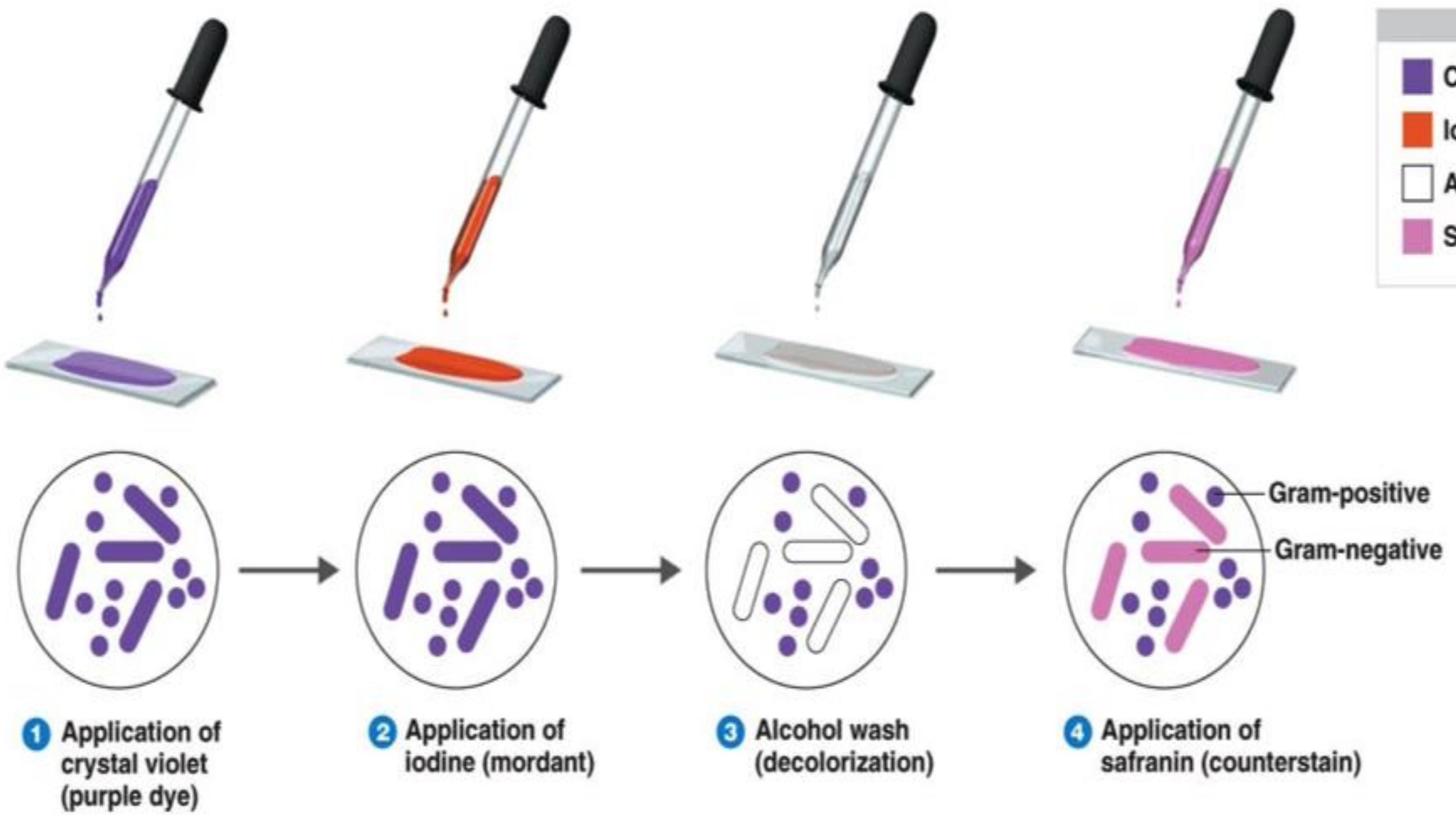
3. DECOLORIZER
   - Acetone + Methanol
   - Removes Crystal violet iodine complex from thin peptidoglycan layers
   - Dissolves outer layer of Gram negative org

4. GRAM SAFRANINE

- Counter stain
- Red colored
- Stains thin walled Gram negative organism



Steps to Gram Stain



Acid-Fast Stain (Ziehl-Neelsen Stain)

1. Air dry and heat fix a thin film of microorganisms.

2. Flood the slide with Carbolfuchsin. Steam the slide with a Bunsen burner. Let the slide set for 5 minutes. Rinse with water.

3. Flood slide with Acid Alcohol for 30 seconds. Rinse with water.

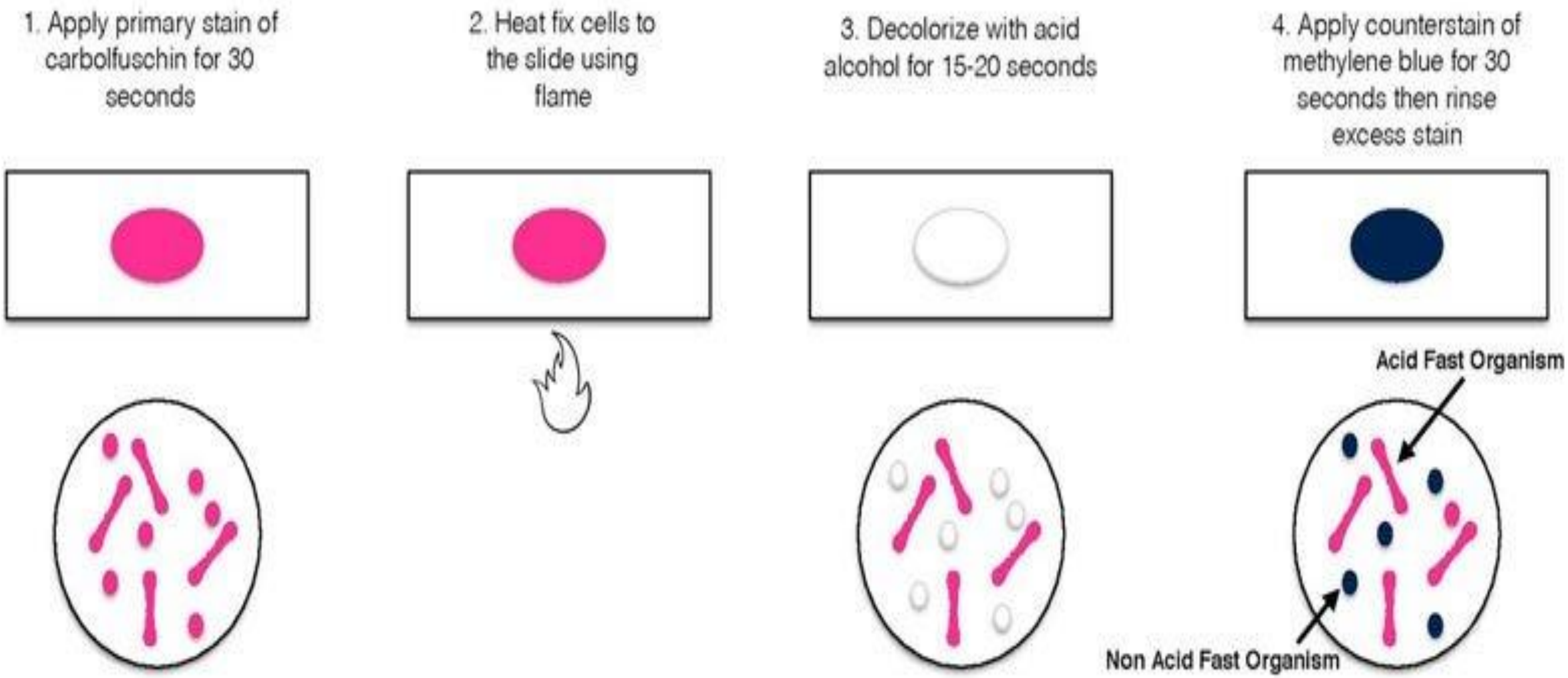
4. Counterstain by flooding the slide with Methylene Blue for 30 seconds. Rinse with water.

5. Dry the slide

6. View organisms using the oil immersion objective of your microscope.

A B C

Procedure for Ziehl-Neelsen Staining



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- https://www.youtube.com/watch?v=EcPriPnjnDw